Too big to lose - a FAIR repository for biomedical data derived from Home-Cage Monitoring

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Abstract

Home-Cage Monitoring (HCM) systems have revolutionised the collection of continuous behavioural data in rodent research and offer unprecedented insights into animal behaviour, yet the full potential of these data remains largely untapped. This paper, developed within the European network COST Action TEATIME, argues for a collaborative approach to harness the power of HCM data through interdisciplinary partnerships with machine learning experts and data scientists. We identify challenges that persist in fully capturing and analysing the complete behavioural repertoire of rodents over extended periods and propose the development of FAIR-compliant (Findable, Accessible, Interoperable, and Reusable) repositories for HCM (meta)data. Standardised data formats and comprehensive metadata enable confident predictions and discoveries across diverse scientific domains. By facilitating data sharing and collaborative analysis, we aim to unlock novel insights into health and disease, ultimately enhancing research efficiency and reducing animal use. Establishing this infrastructure will not only advance our understanding of rodent behaviour but also pave the way for innovative applications of machine learning and big data analytics in biomedical research

Introduction

Recent advancements in artificial intelligence (AI), particularly in machine learning algorithms, have catalysed groundbreaking progress across various scientific fields. The revolutionary impact of these technologies is exemplified by notable achievements like the 2024 Nobel Prize in Chemistry for computational protein design and prediction[1,2], or more recently the development of a foundation model of the mouse visual cortex that can predict dynamic neuronal responses across multiple visual areas, marking significant progress toward an accurate functional digital twin of the mouse visual system[3]. At the heart of this AI-driven revolution is the ability to leverage large datasets. For these datasets to reach their full potential, they must be curated and structured in ways that enable seamless sharing, synthesis, and integration with AI-driven solutions.

In the realm of animal behaviour and biomedicine, technologies such as Home-Cage Monitoring (HCM) systems have made it possible to continuously collect large datasets from laboratory animals, operating 24/7 and over the entire experimental life of an animal[4–6]. While these systems present an opportunity to amass invaluable data, the absence of standardised data formats and the fragmented storage of datasets across inaccessible locations have led to underutilisation of this wealth of information. Furthermore, any data that is shared today is likely to be missing critical metadata—for example, environmental conditions such as ambient temperature and light levels—hindering the ability to merge and integrate datasets across labs, thus limiting the reproducibility of research and stifling innovation.

This “Big Data” challenge is not unique and has been faced, and addressed successfully, in other fields. A recent review of the use of datasets such as The Cancer Genome Atlas (TCGA) in cancer biology described numerous “key big-data analyses that led to conceptual advances in our understanding of cancer biology and impacted disease diagnosis and treatment decisions” [7–10]. Assembled as a way of dealing with the petabytes of data produced by labs across the world, TCGA dataset has contributed to over 10,000 papers in just over 15 years.

At the same time, however, certain fields, such as that of Alzheimer's disease (AD) research,

have seen limited clinically significant breakthroughs despite decades of research. Until recently, all registered AD drugs had been symptomatic treatments, with the last one registered in 2003. In 2021, the first amyloid-β (Aβ) antibody treatment was registered; however, this novel class of drugs is plagued with severe side effects and modest efficacy[11,12]. Despite extensive research into the amyloid cascade and numerous other pathophysiological mechanisms of AD in a "bottom-up" approach, understanding the interplay between even a few such factors is a formidable challenge[13,14]. Though a similar approach has yielded success in cancer research, the aetiology of AD remains largely unknown, limiting the potential for developing safe and effective treatments to modify the neurodegenerative course of the disease[15].

A fruitful “top-down” approach in this example is likely to involve a big-data approach to behaviour. If we posit behaviour as the cumulative output of the brain's cellular and molecular (patho)physiology[16,17], a thorough understanding of a disease as complex as AD may come from a deep and comprehensive analysis of common behavioural patterns and their relationship to the disease's pathophysiology. Indeed, certain behavioural patterns, such as hyperactivity and ADHD-like (Attention Deficit/Hyperactivity Disorder) behaviour, seem to consistently appear in AD models from Drosophilae to rats, but these relationships remain insufficiently explored[18–20]. Given the complexity of both AD and behaviour in general, this avenue of exploration can hardly be pursued without amassing standardised, granular, and longitudinal behavioural data, as is being obtained with the HCM approach. Therefore, to approach significant challenges in neuroscience and neurology through the lens of behaviour, one of the critical missing components is a tailored data repository.

In this position paper, we first use a theoretical and experience-driven approach to develop a structured definition of the requirements for a data repository for HCM experiments. We use this definition to assess whether existing platforms meet the needs of the HCM community. We conclude that there is an urgent need for a bespoke solution, and we propose how this should be structured and explore the various factors that demand attention during such an endeavour.

Methods

Defining Home-cage Monitoring data repository requirements

To stimulate progress and aid in the creation of an HCM data repository, we have used a categorical framework (i.e., an olog, short for ontology log) to define the structure and components of an ideal repository [Figure 1]. Ologs are graphical, category-theoretic tools for knowledge representation that capture entities and relationships in a given domain[21]. By creating an olog for "Home-Cage Monitoring," we mapped out the fundamental components of HCM experiments – including the subjects (animals and their characteristics), environment (enclosures and conditions such as food, water, enrichment), hardware (sensors, actuators, and recording devices), software (data acquisition and analysis algorithms), and the data outputs (behavioral and physiological readouts, timestamps, etc.). This formal representation ensured that we captured the multi-faceted nature of HCM and helped identify key metadata elements required to describe any HCM dataset (e.g., animal strain/age/sex, enclosure conditions, sensor modalities, sampling rates, experimental manipulations, time span of monitoring, etc.). These metadata elements align with recent proposals for minimal metadata in *in vivo* studies, such as the Minimal Metadata Set (MNMS) which emphasises essential descriptors to make preclinical data reusable and FAIR-compliant[22].

Our definition of a suitable HCM data repository is a FAIR-compliant, modular, and federated platform designed to store, index, and enable

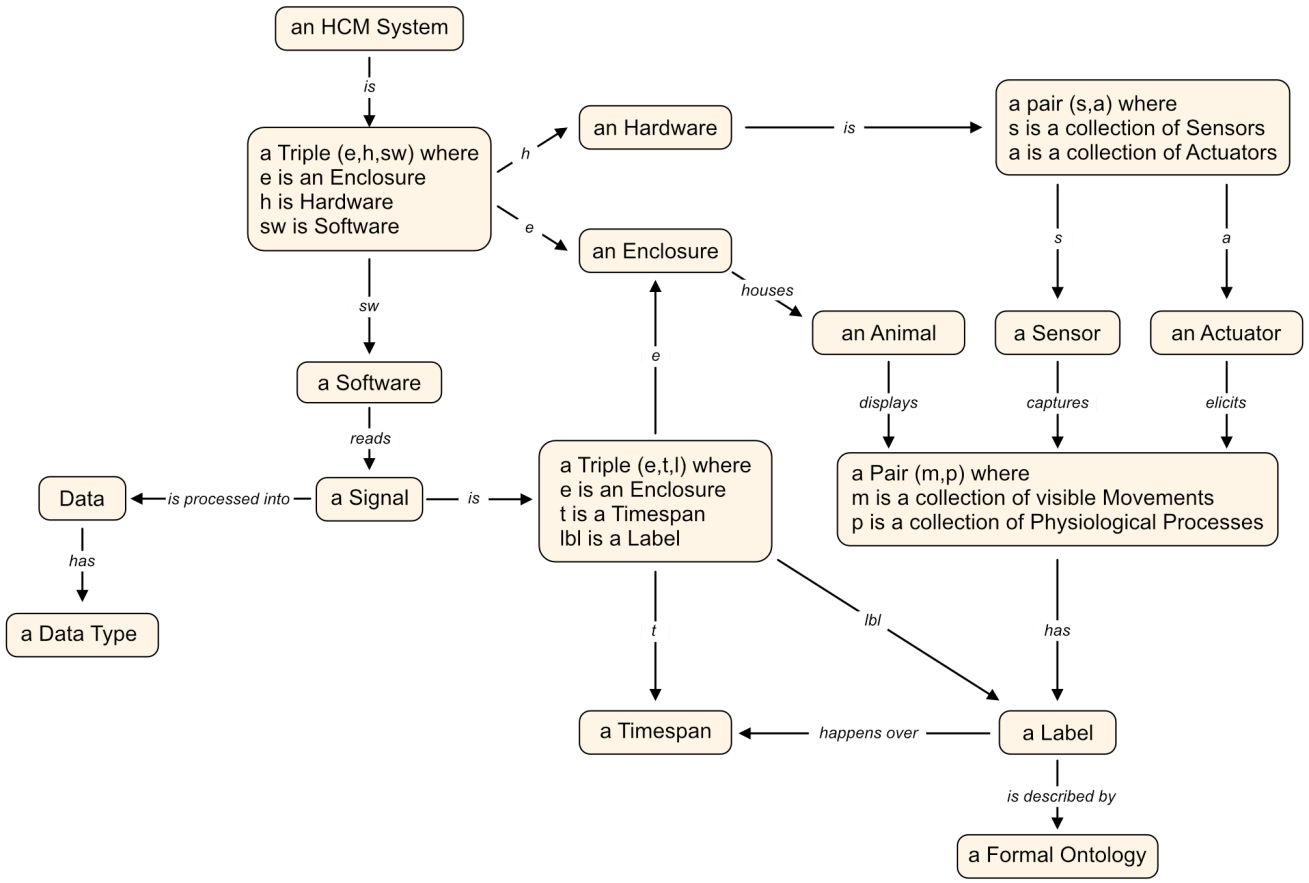

Figure 1. An ontology log (olog) defining the structure and components that are fundamental to home-cage monitoring experiments. Items in boxes represent the various components with connecting arrows describing the relationships between items. The olog ensures that appropriate metadata are collected and associated with raw data in the proposed repository.

analysis of rich, multimodal behavioural datasets collected continuously from laboratory animals. It is designed to strictly follow the FAIR principles[23,24] in order to ensure that all resources within the repository are FAIR. The HCM data repository must support time-series data, video recordings, and flexible metadata describing experimental procedures, animal subjects, and environmental conditions. Unlike generic repositories, it must integrate standardised behavioural ontologies, support machine-readable formats for different use cases (ontology-driven, web-native, legacy-system) such as RDF, JSON-LD, XML, and offer an application programming interface (API) for automated data exchange. Its architecture must enable local transactions, while connecting globally via a searchable portal, promoting reusability, transparency, and cross-study comparison. Critically, it will bridge academic and industry standards, such as SEND (Standard for Exchange of Nonclinical Data), facilitating regulatory alignment and translational research[25,26].

This framework builds on initiatives like VICT3R[27], which promotes virtual control groups to reduce animal use, and CEDAR[28,29], a metadata platform integrated with OSF[30] and compatible with BioPortal ontologies[31]. It aims to harness the power of Big Data analysis and machine learning to transform biomedical research while fostering collaboration among

stakeholders in animal research to uncover novel insights that will drive future breakthroughs.

Evaluation of Existing Data Repository Platforms

Having established the requirements, we conducted a structured evaluation of existing data repository platforms to determine how well they fulfil our defined criteria. Our assessment focused on systems capable of handling three critical data types:

- Flexible metadata for experimental design, subjects, and conditions
- Time-series data from diverse sources (e.g. environmental parameters, animal activity, RFID readings)
- Video footage for behavioural recognition and pattern analysis

Additionally, we sought systems with an API to facilitate automated data collection and transfer.

Our evaluation methodology combined literature review, online resources, and hands-on exploration of each platform. We reviewed documentation and user guides to ascertain technical capabilities (storage limits, supported metadata standards, API availability, etc.), and when possible, we uploaded test datasets or used demo instances to assess usability. We used the olog-defined use cases as test scenarios to probe each platform's fitness.

The evaluation process included:

- Initial web search: identified 49 candidate repositories
- Preliminary screening: narrowed the list to 16 systems during a dedicated TEATIME meeting at the Mary Lyon Centre at MRC Harwell, UK
- Scoring criteria: evaluated systems based on data searchability, video data storage and playback capabilities, and public accessibility
- In-depth testing: conducted using synthetic behavioural recordings on 13 selected databases

The following databases underwent final evaluation: Figshare, Dryad, Open Science Framework (OSF), Science Data Bank (ScienceDB), Dataverse (Harvard implementation), Synapse, BioStudies, Zenodo, Mendeley Data, Databrary, GigaScience Database (GigaDB), B2SHARE, dbGaP. Details of the original list of repositories can be found in Supplementary Table 1.

Sociotechnical Considerations for HCM Repository Implementation

Practical implementation requires addressing the sociotechnical landscape in which researchers operate. Current data-sharing practices are shaped not only by technical capabilities but also by institutional policies, research culture, resource constraints, and professional incentives. Achieving widespread adoption of a FAIR-compliant HCM data repository therefore requires strategies that acknowledge these real-world considerations[22,32,33].

Flexibility and Inclusiveness

For the repository to be truly impactful, it needs to be able to accept contributions from all laboratories and experiments, up to and including large-scale and long-term studies. This means designing standards that accommodate diverse research contexts and resource levels; a layered approach to metadata and data requirements is therefore essential for the HCM repository framework. This includes mandatory elements that provide core information for basic interoperability (e.g., species, strain, experimental duration, sampling frequency), recommended elements

representing best practices that enhance reusability (e.g., detailed environmental conditions, animal housing specifications), and optional elements offering additional details that enrich the dataset for specialised analyses (e.g., custom behavioural classifications, protocol variations). This balanced approach makes the repository's structure flexible and inclusive, ensuring accessibility for all contributors while supporting comprehensive data documentation.

Contributions would be encouraged from laboratories that can only meet core requirements, as well as from well-equipped facilities that can provide richer contextual information. The tiered structure would create an accessible entry point for all contributors while establishing a pathway toward more comprehensive data documentation as capacity develops[22].

The challenge of behavioural terminology illustrates a flexible approach in practice, as behavioural research employs diverse classification systems ranging from traditional ethograms to novel computational metrics. Instead of enforcing a single classification scheme, the repository should adopt a dual strategy by supporting and/or developing a controlled vocabulary derived and adapted from existing ontologies (like the Neuro Behavior Ontology[34] (http://purl.obolibrary.org/obo/nbo.owl) to facilitate cross-study comparison, while also allowing custom classifications with optional mapping to standard terms where applicable (similar to CEDAR metadata schemas and BioPortal integrations). This balanced approach ensures consistent annotation across studies without excluding specialised measures developed for unique experimental paradigms or innovation.

Incentives and Recognition for Data Sharing

Even with flexible standards and respect for local autonomy, widespread adoption remains unlikely without addressing the fundamental incentive structures of academic research. Current academic culture primarily rewards publication in high-impact journals, with limited recognition for the substantial effort involved in data curation and sharing. This misalignment creates a situation in which researchers perceive data sharing as additional work with minimal professional benefit. Studies across disciplines have identified lack of recognition as a major barrier to open data sharing[35]. Importantly, acknowledging all contributors in a granular fashion promotes a more collaborative ethos and incentivises the often labour-intensive work of preparing high-quality, reusable data.

To address this, we propose to integrate a formal contribution recognition system into the repository architecture. Drawing inspiration from the CRediT (Contributor Roles Taxonomy) system used in publications, and from the data-index valuing data sharing[36], the repository would include a structured framework to enumerate the roles played by each person in the data lifecycle. These should include, but not be limited to, the following roles: Data acquisition, Experimental setup and maintenance, Animal handling and care, Protocol design and optimization, Data curation and annotation, Software development, Data validation and quality control, Project supervision and coordination.

By linking these contribution records to persistent identifiers (such as ORCID (Open Researcher and Contributor ID)), we can ensure individuals receive public credit for their work throughout the research ecosystem. The repository could implement this through technical features like contribution statements embedded in metadata records, automated generation of citations for dataset usage, integration with ORCID profiles to track data contributions, and metrics that quantify dataset reuse and impact.

The acknowledgment of data contributions through a dedicated system would greatly

advantage early-career researchers, technicians, and data specialists, as their crucial work frequently goes unnoticed in traditional publication models. Implementing data portfolios for researchers would create a new type of scholarly output that hiring committees and funders could recognise if properly visible.

Bridging Local and Global Initiatives

A significant challenge in research data management is the current disconnect between local and global initiatives. Many institutions have invested and developed isolated solutions for HCM data—ranging from IT-department-led platforms to researcher-driven ad-hoc systems like spreadsheets and custom scripts. These local solutions typically reflect specific research workflows, institutional policies, and available resources. While functional within their context, these isolated systems create data silos that hinder broader discovery and reuse[37].

The proposed repository framework must address this fragmentation while respecting institutional autonomy. Many laboratories and institutions are reluctant to transfer control of their raw data to external entities[38,39] for several legitimate reasons such as:

- Governance requirements: institutional policies often mandate local storage and control of research data
- Security concerns: transferring sensitive data to external systems introduces perceived risks
- Infrastructure investment: existing local storage and computing resources represent significant investment
- Workflow integration: research teams have developed processes optimised for their local systems

A repository design that ignores these concerns by demanding centralised data submission would face significant adoption barriers. Instead, we propose a federation strategy that respects local autonomy while enabling global discovery and collaboration.

In a federated architecture, laboratories would maintain primary data storage within their existing infrastructure while participating in a networked ecosystem through metadata sharing and standardised interfaces. Each participating node could implement the repository software locally (through containerisation or virtual appliances), maintaining full control over physical storage location, access permissions and authentication, integration with local systems and compliance with institutional policies.

At the same time, global discovery would become possible through a lightweight indexing layer that harvests metadata from participating nodes. This global index would serve as a searchable catalog without storing or controlling the raw data itself. Researchers could query across the entire network to discover relevant datasets and then access them according to the policies established by each data owner. This design has proven successful in other domains—for instance, the Global Alliance for Genomics and Health (GA4GH) employs a similar federated model for genomic data sharing[37]. In practical terms, a researcher could use the HCM global portal to find datasets of interest (via searchable metadata) and then be directed to the data's home node for access, subject to that node's access policies.

Importantly, this federated approach scales organically from single-laboratory deployments to international networks. A laboratory can initially implement the repository software as a stand-alone system, improving local data management without immediate data sharing. As comfort and capacity develop, they can selectively expose datasets to the global index, gradually increasing participation in the broader ecosystem. This incremental path lowers adoption barriers by allowing participants to engage at a comfortable pace rather than forcing an all-or-nothing decision.

Bringing together these considerations, we envision a modular repository framework that can be deployed across diverse research environments while maintaining interoperability through common standards and protocols.

Results

Evaluation Outcomes of Existing Repository Systems

Our assessment revealed that most existing systems are primarily designed as static repositories for supplementary files associated with specific publications. Key limitations included:

- Lack of integration between online video data and time-series measurements
- Absence of tagging capabilities for manual behaviour recognition and annotation
- Insufficient flexibility for handling diverse HCM data types

Based on our evaluation, we determined that no existing repository system fully meets the specific requirements for HCM data storage for sharing and analysis. Thus, our findings underscore the need to develop a detailed specification for a centralised data hub capable of handling various HCM systems and to establish industry standards for future implementations.

The Need for a Bespoke Solution

One of the main objectives of the COST Action CA20135 *TEATIME* was to identify the advances required in the field of Home-cage Monitoring with the aim of enhancing research coordination and capacity building. Data sharing and effective communication of results sits at the heart of this endeavour. Therefore, the consortium undertook an evaluation of existing biomedical data repositories to determine whether a suitable repository for HCM data exists. Our evaluation [see Supplementary Table 2] revealed that most systems are storage-focused, primarily housing data tied to specific publications as static supplementary files. Systems with online video capabilities lack integration with time-series measurements or manual behaviour annotations. These gaps underscore the need for a centralised platform that incorporates feedback from both data creators and users to drive future developments. A comprehensive repository must incorporate features to handle three key data types:

- Flexible metadata covering experimental design, subjects, and conditions.
- Time-series data from diverse sources, including environmental conditions, animal activity, telemetry, and physiological recordings.
- Video footage storage for behavioural recognition, enabling AI-driven or manual analysis alongside metadata and time-series data. In addition, the repository should include an application programming interface (API) to enable automated data collection and seamless uploads.

To address these challenges, the COST Action CA20135 *TEATIME* advocates for the creation of a FAIR-compliant open repository for HCM data and metadata. Such a repository would pave the way for integrating AI and Big Data analytics with biomedical research, enhancing our understanding of the links between behaviour and disease. By integrating detailed behavioural records with metadata and environmental conditions, this initiative would facilitate comprehensive exploration of behavioural patterns and their links to disease outcomes using AI. Furthermore, it would reduce the need for new animal experiments by enabling novel hypotheses to be tested with existing datasets, thereby advancing animal welfare and adhering to the 3Rs (Refinement, Reduction, Replacement) principles[40] .

A Framework for Building a FAIR-compliant HCM Data Repository

The adoption strategy for the HCM repository hinges on meeting researchers where they are. By being flexible in standards (inclusive of small datasets and custom behaviors), respecting local controls through a federated design, and actively incentivising data sharing via contributor recognition, we can dramatically lower socio-technical barriers. By implementing these components as a modular, standards-based system, we can create a framework that laboratories can adopt incrementally, according to their resources and needs. These enhancements would serve the broader goals of transparency and collaboration, whereby researchers could contribute data without excessive burden, retain credit and ownership, and in turn benefit from access to a richer pool of shared data.

Throughout all layers, the proposed repository framework or architecture implements FAIR principles[23] through specific technical choices:

- Findable:
  - Each dataset would be assigned a persistent identifier (e.g., DOI), ensuring long-term discoverability.
  - Rich, structured metadata, conforming to standardised schemas, would be indexed in the global discovery portal.
  - The global portal would provide a comprehensive search interface, allowing users to find datasets based on various criteria (e.g., species, strain, experimental conditions, behavioural parameters).
- Accessible:
  - Data would be accessible via standard HTTP/HTTPS protocols, ensuring broad compatibility with web browsers and other applications.
  - Flexible authentication and authorisation mechanisms (e.g., OAuth 2.0, RBAC) would be implemented to manage data access while respecting data owner policies.
  - APIs (RESTful or GraphQL) would be available for programmatic access to data and metadata, facilitating automated data retrieval and analysis.
- Interoperable:
  - Common data formats (e.g., JSON, CSV, RDF, OWL) would be supported to ensure data can be exchanged and processed by different software tools.
  - Semantic technologies (e.g., OWL ontologies, RDF knowledge graphs, SPARQL queries) would be used to represent data and relationships, enabling sophisticated data integration and analysis.
  - Mappings to regulatory standards (e.g., SEND) would be provided to facilitate data exchange and compliance in translational research.
- Reusable:
  - Comprehensive metadata would be provided, including detailed descriptions of experimental design, methods, and data provenance, ensuring that data can be interpreted and reused appropriately.
  - Clear data usage licenses (e.g., Creative Commons) would be specified, indicating the conditions under which data can be reused.
  - Data dictionaries and codebooks would be provided to document the meaning and structure of data elements.
  - Contribution statements and ORCIDs would be used to give credit to data creators and stewards, promoting a culture of attribution and reuse.

This architecture would directly address the requirements while accommodating the socio-technical considerations [see

Supplementary Table 3]. Ultimately, this approach would not only facilitate data reuse and integrative analysis in Home-Cage Monitoring, but also help build an open community of practice, in which sharing data is rewarded, and would lead to new discoveries. This framework is a practical path forward that reconciles real-world constraints with the ideals of open science and FAIR data stewardship.

Technical Architecture of the Proposed Repository

The proposed example architecture employs a modular, federated approach composed of an underlying API-first framework and four integrated layers designed to meet the specific requirements of an HCM data repository. This framework would act as the central node of the repository, facilitating communication, data exchange, and integration between all components [Figure 2].

For example, the API platform framework (https://api-platform.com/) offers a unified interface for the HCM repository, ensuring consistency, security, and scalability through features like standardised RESTful or GraphQL API design, robust authentication and authorisation, API gateway management for traffic and versioning, and comprehensive monitoring and logging tools, as well as interactive documentation using Swagger/OpenAPI. This framework would streamline development and maintenance by providing a reliable foundation and would promote interoperability by standardising communication protocols between all repository components.

Local Repository Nodes

The foundation of the system would be a containerised application that laboratories can deploy locally with minimal IT support, gradually migrating existing data into the structured framework at a manageable pace. Using technologies like Docker (Accelerated Container Application Development) would enable consistent functionality across heterogeneous computing environments while respecting local infrastructure constraints. Each node would include:

- Standardised Metadata Schemas: implemented in machine-parsable formats such as JSON Schema or XML Schema. These schemas would define the structure and validation rules for metadata related to experiments, subjects, conditions, and data files.
- APIs for Data Ingestion, Query, and Retrieval: RESTful (or GraphQL) APIs would be developed to facilitate automated data exchange. These APIs would support operations like uploading time-series data, video files, and metadata, as well as querying and retrieving (meta)data based on various criteria (e.g., date range, subject ID, behaviour type).
- Data Storage: each node would manage its own data storage, which could be a local file system, a network-attached storage (NAS), or a cloud storage service. The system would support varied data formats, including CSV (or NWB or H5F) for time-series data, MP4 for videos, and JSON for metadata.
- User Interfaces (UIs) for Data Curation and Management: web-based UIs would be provided for manual data entry, metadata editing, and data management tasks. These UIs would be designed to be user-friendly and accessible to researchers with varying levels of technical expertise.
- Access Control Mechanisms: role-based access control (RBAC) would be implemented to manage data access permissions. This would allow each node to control who can view, edit, or download data, ensuring compliance with institutional policies and data privacy regulations.

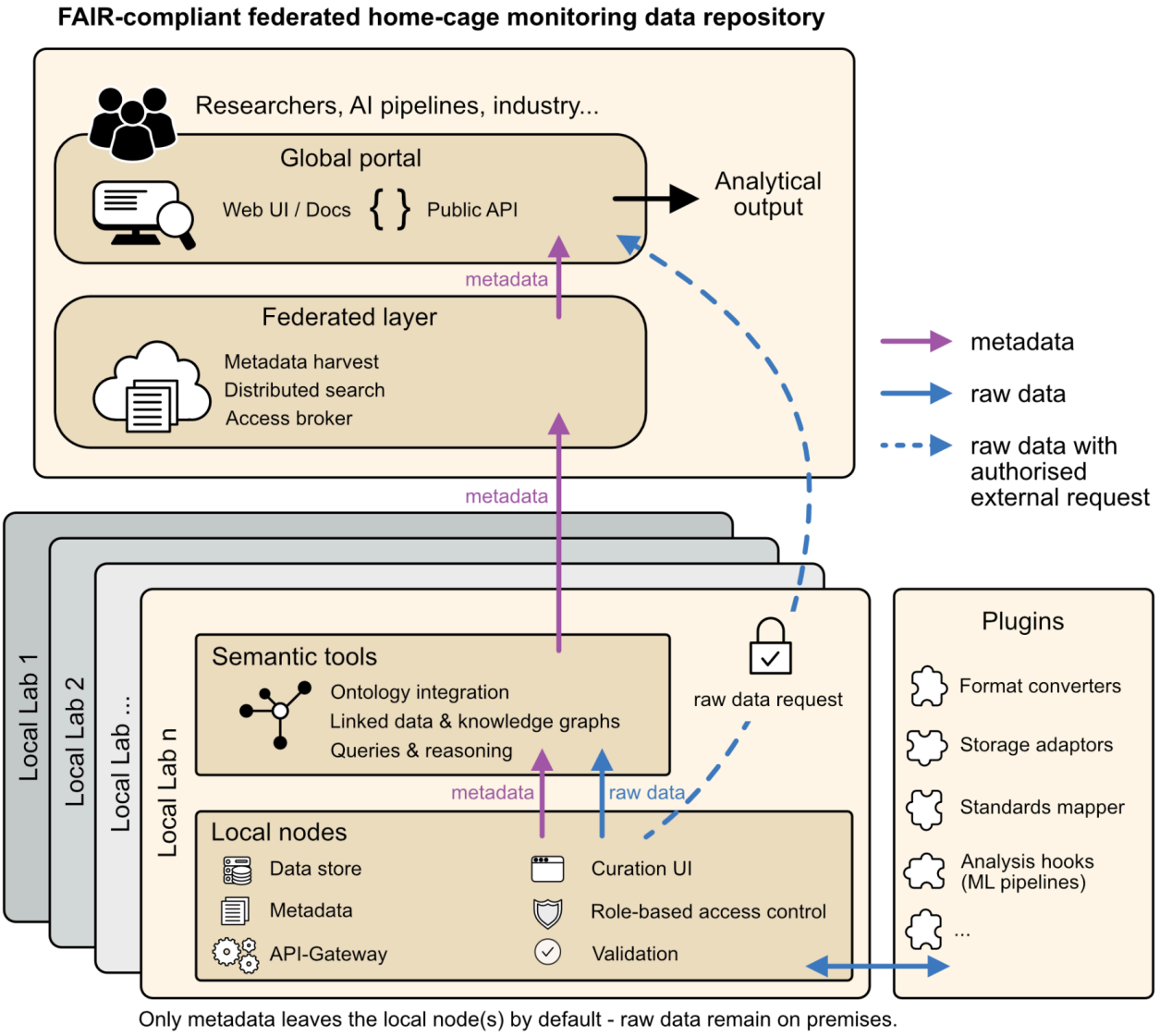

Figure 2. Layered, federated architecture of the proposed home-cage monitoring data repository. Local lab nodes (bottom panels) retain raw data but expose rich metadata (purple arrows) through standardised interfaces. Semantic tools harmonise vocabularies via ontology integration. A federated layer (top panel) harvests metadata for distributed search and access brokering. A global portal delivers findability and programmatic access for various stakeholders including academic researchers, artificial intelligence pipelines, and industry. Interoperability plugins enable connection with existing research infrastructure through storage adapters, format converters, standards integration, and analysis pipeline connections.

- Data Validation: upon ingestion, data would be validated against the defined schemas to ensure data integrity and consistency. Any invalid data would be flagged and rejected or quarantined for further review.

This localized deployment would respect institutional autonomy and data sovereignty while providing standardised functionality. Laboratories could implement the repository software with minimal IT support, gradually migrating existing data into the structured framework at a manageable pace.

Semantic Data Layer

To enable sophisticated data integration and discovery, the repository would implement a

semantic representation layer using established web technologies[41–45] such as:

- Ontology Integration: the system would incorporate existing behavioral ontologies (e.g., Neuro Behavior Ontology) and extend them where necessary for HCM-specific concepts and user-specific concepts. This would provide a controlled vocabulary for consistent annotation across studies. Ontologies would be represented using the Web Ontology Language (OWL), which is based on the Resource Description Framework (RDF), and serialised using a syntax such as Turtle.
- Knowledge representation: data relationships would be encoded using RDF (Resource Description Framework) or JSON-LD (JSON for Linked Data) formats, creating machine-parsable knowledge graphs that express connections between experimental elements (subjects, behaviors, conditions). For example, this would enable a specific behavioural observation to be linked to environmental conditions, genetic background, and temporal context. Each piece of data would be represented as a "triple" (subject-predicate-object), forming a network of interconnected information.
- Semantic queries: with structured semantic data, researchers could perform sophisticated cross-dataset SPARQL queries (using the SPARQL language) that would be impossible with traditional databases. An example would be the identification of all datasets showing altered circadian rhythms in a specific genetic model across multiple laboratories: with a SPARQL query, a researcher wanting to find all studies in which a particular gene mutation resulted in changes to the sleep–wake cycle in mice, could ask the system to retrieve only the datasets that match these very specific criteria, even if the data is spread across different storage locations.
- Reasoning and Inference: the semantic layer could also support reasoning and inference, allowing the system to automatically derive new knowledge from existing data. For example, if the ontology defines that "increased activity" is a sign of "stress," the system could support basic rule-based inference (like activity threshold → stress), with the potential for more sophisticated machine-learning approaches later. Rather than AI-generated conclusions, it would provide structured relationships between behaviours and internal states that could be validated by researchers. The value lies in systematically capturing and leveraging these relationships across datasets.

This semantic approach would provide the technical foundation for meaningful data integration while maintaining sufficient flexibility for specialised research domains. By leveraging established semantic web standards, the repository would benefit from existing tools and expertise while ensuring future compatibility with evolving technologies.

Federation Mechanism

The distributed nature of the system would be managed through a federation layer that connects local instances:

- Metadata Harvesting: the global portal would periodically collect metadata from participating nodes using modern APIs, including RESTful or GraphQL interfaces, and potentially established protocols like OAI-PMH for archival purposes.
- Distributed Search: queries submitted to the global portal would be distributed across the network, with results aggregated and returned to the user. The portal would use techniques such as federated search or distributed query processing to efficiently search across multiple nodes. For example, if each local node exposes its metadata via a SPARQL endpoint (a standardised way to query

RDF data), the global portal could use a federated SPARQL query to simultaneously query all participating nodes. The portal would then collect the results from each endpoint, combine them, and present a unified set of findings to the user. This approach would allow for complex semantic queries across the entire network, enabling researchers to find relevant data even if it is spread across many different locations.

- Access Brokering: when a user discovers a relevant dataset via the global portal's federated search (powered by SPARQL queries against RDF data exposed through local SPARQL endpoints), the system would facilitate access according to the policies set by the data owner. This might involve direct API connections using RESTful or GraphQL APIs, or through authorised download mechanisms. The portal would provide links to the data's home node, where users would request access and download the data. Authentication and authorisation would be managed according to the Role-Based Access Control (RBAC) implemented at each local node. The access requests and granted permissions could be logged and tracked using JSON-LD metadata to maintain a clear audit trail and ensure compliance with data governance policies.

Interoperability Plugins

A plugin architecture would enable integration with existing research infrastructure:

- Storage Adapters: connectors for various storage backends (institutional servers, cloud services, general repositories) would allow flexibility in data management. These adapters would handle the specific details of interacting with different storage systems, such as authentication, file transfer, and data retrieval.
- Format Converters: tools to transform data between common formats would ensure interoperability with analysis software and other repositories. For example, a converter could transform CSV data to JSON or vice versa.
- Standards Integration: mappings to regulatory standards like SEND would facilitate the translational pathway from discovery to application. Plugins would be developed to ensure that data can be exported in SEND format, which is required for regulatory submissions.
- Analysis Pipeline Connections: APIs that enable seamless integration with automated analysis tools and machine-learning frameworks. These APIs would allow external tools to access and process data from the repository, enabling automated analysis and data mining.

By providing connectors (plugins) for common platforms, the framework would avoid locking users into a specific storage mode and would make integration into existing workflows easier. Likewise, for the global portal, plugins or adapters could handle interfacing with the diverse local nodes—translating between the portal's query language and each node's API or database schema. This modular approach, much like adapters in distributed databases, guarantees that adding a new node or connecting to a new storage system would not require rewriting the whole platform, only developing a new plugin module if one does not already exist.

Discussion

The need for a comprehensive, centralised platform for animal behaviour and biomedical data has never been more urgent. As highlighted by the preliminary evaluation [see Supplementary Table 2] conducted during COST Action TEATIME, current biomedical data repositories tend to be storage-centric, offering static data storage, fragmented video integration, and insufficient metadata frameworks.

These limitations hinder the potential for robust analysis and the synthesis of diverse data types necessary for advancing HCM research. We propose the creation of a FAIR-compliant repository [see Supplementary Table 3] tailored to the unique needs of HCM technologies. Such a repository would enable seamless integration with AI-driven analysis tools by unifying experimental metadata, time-series measurements, and video datasets under a scalable cloud infrastructure. Such a platform would consolidate key data types—flexible metadata, time-series data, and video footage—ensuring that diverse datasets can be effectively shared, analysed, and reused.

Large-scale biomedical research data repositories have revolutionised data sharing, usage and analysis, with real-world consequences. Whether by enabling pharmaceutical companies to de-risk drug targets by mapping genotypes and phenotypes for drug target validation and creating risk prediction models[46–48] as in the case of the UK Biobank, or by enabling targeted cancer therapies through the integration of whole genome sequencing data of patients into clinical diagnosis, as in the case of the UK's National Health Service (NHS) through the 100,000 Genomes Project UK[49,50], such initiatives herald a new era of tailored therapies for some of the most devastating conditions touching human lives.

Building on these strengths, our proposed framework [see Figure 2] emphasises data ownership and researcher attribution within a community-driven model. Unlike generic repositories, we propose a custom solution for HCM research that is iteratively curated by a multidisciplinary scientific community, establishing a robust, standardised infrastructure. This collaborative approach would minimise redundancies, accelerate biomedical discovery, and elevate animal welfare practices by enabling more efficient use of data and reducing the need for new animal experiments.

A repository built on such a framework has the potential to drive transformative changes not only in biomedical research but also in societal, technological, and economic landscapes. By creating a centralised platform, such a repository would break down barriers to interdisciplinary collaboration, ultimately advancing therapeutic innovations and facilitating the translation of animal research findings into clinical applications. Such a repository would empower researchers, AI communities, and industry partners to collaborate more effectively, driving the development of new treatments and technologies while maintaining a strong ethical foundation rooted in animal welfare principles.

For researchers, a repository based on such a framework represents a transformative advance in preclinical and basic science. By delivering standardised, reproducible workflows, the reliability, scalability, and translational value of rodent behavioural studies would be significantly enhanced. With deeper, structured insights into animal models, the platform would accelerate the journey from bench to bedside—helping ensure that robust preclinical findings inform clinical trials and new therapies, as demonstrated by landmark initiatives such as TCGA, the UK Biobank and the 100,000 Genomes Project UK[46–50].

Two key enablers of this impact would be the adoption of a controlled vocabulary that promotes precise, cross-disciplinary communication around behaviour and disease phenotypes, and a layered approach to data entry. In this way, a flexible but principled approach would be fostered, ensuring semantic consistency across studies without stifling innovation or excluding specialised methodologies, while still upholding the principles of the 3Rs—Replacement, Reduction, and Refinement—that guide responsible animal research.

For the AI and machine-learning communities, the availability of high-quality, longitudinal datasets would provide fertile ground for the

development of sophisticated models that could potentially uncover previously unrecognised environmental factors, predict comorbidities, and refine disease models[51,52]. In turn, this would advance the predictive power of AI in biomedical research, making it a key tool in identifying novel therapeutic targets and improving the precision of clinical interventions. By leveraging these rich datasets, researchers would be able to explore new patterns and uncover insights that would otherwise have remained hidden, potentially revolutionising how we understand the relationship between behaviour and disease.

Industry stakeholders, including pharmaceutical companies, contract research organisations (CROs), and funding bodies, stand to benefit greatly from the establishment of such a repository. The platform would enhance data integrity, ensuring that preclinical research adheres to the highest regulatory and ethical standards, while also driving innovation in drug development. Long-term storage and accessibility of high-quality datasets would provide a foundation for future breakthroughs, making it easier for commercial entities to identify promising research and guiding investment in areas with the most potential for clinical success.

Moreover, a community-driven repository would support the continuous evolution of standards[53] in animal behaviour research, benefiting from input from experts in data science, machine learning, and related fields. The collaboration facilitated through such a repository would foster the development of cutting-edge tools and methodologies, ensuring the ongoing relevance and improvement of the platform. As a dynamic resource, the repository would adapt to emerging needs, promoting the adoption of best practices and the integration of innovative technologies across the global research community.

Beyond its scientific impact, the repository would serve as an educational tool, training the next generation of researchers in ethical animal research and behavioural phenotyping. By incorporating hands-on learning and real-world examples, it would equip emerging scientists with the knowledge and ethical foundation to shape the future of biomedical research. This, in turn, would promote responsible scientific inquiry and ensure that animal welfare remains at the forefront of research practices.

Ultimately, the creation of such a large-scale data repository for HCM data would bridge gaps between different communities in academia, industry, and technology, catalysing a paradigm shift in behavioural research. It would help build public trust in animal research by aligning scientific advancements with societal expectations for ethical conduct, transparency, and responsible innovation. By fostering collaboration, reducing redundancy, and enhancing data quality, the proposed repository would unlock immense potential for innovation, ensuring its long-term societal benefit and transforming the future of biomedical research.

In uniting technical innovation with community-driven design, this HCM Data Repository framework reimagines how behavioural data can be shared, discovered, and used across institutional, regulatory, and national boundaries. Rooted in FAIR principles and designed for inclusivity from the outset, the repository would make data contributions visible, citable, and professionally rewarding—turning data sharing into a meaningful form of scholarly output. By establishing common minimal frameworks and modular standards, it would bridge disparate research environments, including regulated and non-regulated domains, enabling a seamless flow of information from academic discovery to regulatory submission. As demonstrated in clinical domains, global data exchange is possible when built on sustained collaboration, shared models, and intelligent querying frameworks[9,54,55]. Our proposed approach mirrors these proven strategies and aligns with ongoing industry initiatives, positioning HCM

technology as a key enabler of interoperable, cross-sector translational research. With committed collaboration from academia, industry, and funding bodies, the repository stands to transform data into a durable, actionable asset—accelerating scientific progress while honouring the full spectrum of contributions that make it possible.

Data availability

The authors declare they did not use any data in the creation of this article. Therefore, no data must be made available.

Code availability

The authors declare that no custom code was used in the creation of this article. Therefore, no code must be made available.

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Contributions

Conceptual Problem Framing: RSB, JR, JEM, PMB, MCA, LL, DV, LR, DH, BG, OK, HF, MR, HG. Ontology & Semantic Modeling: DV, LR. Repository Analysis and Evaluation: PMB, RSB, JR, JEM, MA, LL, DV, LR, OK. FAIR Compliance Specification: DH. Sociotechnical Strategy & Adoption Pathway: BG. Federated & Layered Architecture Design: BG. Technical Infrastructure Coherence: DH. Literature Review & Background Research: RSB, JR, JEM, PMB, MCA, LL, DV, LR, DH, BG, OK, HF, MR, HG. Writing – Original Draft: BG, DH, RSB, JEM, DV. Writing – Review & Editing: JR, PMB, MCA, LL, LR, OK, HF, MR, HG. Visualization & Supplementary Material: JEM, DV, LR, BG, RSB, PMB, DH. Project Administration & Coordination: RSB, DH, BG. Funding Acquisition: HG, JR, RSB. All authors read and approved the submitted version of the manuscript.

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Ethics declarations

Competing interests

Damien Huzard is the owner of NeuroNautix (registered as sole proprietorship in France) and the creator of Metadatapp, a metadata management solution. While these did not fund or contribute directly to the research presented here, the work aligns with their scope of activity and may indirectly increase their visibility.

All other authors declare no competing interests.

Supplementary Tables Legends

Supplementary Table 1. List and Description of Repositories. This table presents the 49 original data repositories shortlisted according to search criteria during the COST Action TEATIME Working Group meeting at the Mary Lyon Centre at MRC Harwell in October 2024. These repositories were initially identified as potential candidates for hosting Home-Cage Monitoring data and represent a comprehensive overview of existing biomedical data storage solutions evaluated by the consortium.

Supplementary Table 2. Evaluation of Repositories. This table details the evaluation results for the shortlisted repositories, providing the rationale behind inclusion or exclusion from the final list of 16 repositories that underwent comprehensive testing. The scoring criteria assessed each repository's capabilities for handling flexible metadata, time-series data, and video footage—the three critical data types identified as essential for Home-Cage Monitoring research. This evaluation directly informed the consortium's conclusion regarding the need for a bespoke repository solution.

Supplementary Table 3. Implementation of FAIR Principles in a Home-Cage Monitoring Data Repository. This table presents a detailed mapping of FAIR (Findable, Accessible, Interoperable, and Reusable) principles to specific implementation examples for a Home-Cage Monitoring (HCM) data repository. Each row illustrates how a particular FAIR principle can be practically realized within the proposed repository framework, ensuring that HCM datasets are properly indexed, accessible, compatible with existing standards, and thoroughly documented for reuse across the scientific community. These implementations are integrated in the federated architecture described and provide concrete examples of how the repository would adhere to established data management standards while addressing the unique requirements of continuous behavioral monitoring data.